

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 8

2024

1. Prove or disprove:

- (a) If (x_n) is a Cauchy sequence and (x_n) and (y_n) are neighbours, then (y_n) is Cauchy.
- (b) If $f : X \rightarrow Y$ is a continuous function between metric spaces and (x_n) is a Cauchy sequence in X , then $(f(x_n))$ is a Cauchy sequence in Y .

2. (a) If (x_n) is a sequence in a metric space (X, d) for which $d(x_n, x_{n+1}) \rightarrow 0$ as $n \rightarrow \infty$, you cannot automatically conclude that (x_n) is Cauchy. Show this by considering, in the reals with the usual metric, the sequence

$$x_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}.$$

You need not provide lots of working; the crucial facts ought to be familiar from real analysis. (Please read this carefully: x_n is NOT $\frac{1}{n}$.)

- (b) Prove that, in an ultrametric space, a sequence (x_n) is a Cauchy sequence iff $d(x_n, x_{n+1}) \rightarrow 0$.

3. If d_D is the discrete metric on the set X , prove that (X, d_D) is a complete metric space. (Hint: Show that all Cauchy sequences in this space are eventually constant.)

4. We call P a **topological property** if it satisfies this condition: whenever a metric space X has property P and X is homeomorphic to a metric space Y , then Y also has property P .

Show that completeness is not a topological property, by considering the metric spaces (X, d_E) and (X, d_D) where

$$X = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$$

and d_E is the usual metric of the reals and d_D is the discrete metric.

5. Let A and B be subsets of a metric space (X, d) . Prove that, if A and B are complete subsets of X , then $A \cap B$ is a complete subset of X .

6. Read the lemma and theorem below, and answer the questions that follow.

Lemma Suppose that (x_n) is a Cauchy sequence in a metric space (X, d) , and $x \in X$. Also suppose (x_{n_k}) is a subsequence of (x_n) such that $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$. Then $x_n \rightarrow x$ as $n \rightarrow \infty$.

Proof: Let $\epsilon > 0$ be given. (1)

Choose $N_1 \in \mathbb{N}$ such that $k \geq N_1$ implies that $d(x_{n_k}, x) < \frac{\epsilon}{2}$. (2)

Choose $N_2 \in \mathbb{N}$ such that $m \geq n \geq N_2$ implies that $d(x_m, x_n) < \frac{\epsilon}{2}$. (3)

Let N be the maximum of N_1 and N_2 . (4)

Now $d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x)$ for all k . (5)

Using this, we get that $n \geq N$ implies that $d(x_n, x) < \epsilon$. (6)

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- (a) How do we know that the N_1 chosen in line (2) exists?
- (b) How do we know that the N_2 chosen in line (3) exists?
- (c) Show how line (6) follows from the previous working. In the process, you should make it clear what value of k you are using.

P.T.O

Theorem If A and B are complete subsets of a metric space (X, d) , then $A \cup B$ is complete.

Proof: Suppose that (x_n) is a Cauchy sequence in $A \cup B$. (1)
 Then (x_n) must have a subsequence in A or in B (or both, of course). (2)
 Suppose without loss of generality that (x_{n_k}) is a subsequence of (x_n) in A . (3)
 Then (x_{n_k}) is a Cauchy sequence in A . (4)
 So $x_{n_k} \rightarrow a$, for some $a \in A$. (5)
 So $x_n \rightarrow a$. (6)
 Since $a \in A \cup B$, this completes the proof. (7)

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- (a) Justify the claim in line (2).
- (b) Explain what the expression “suppose without loss of generality” in line (3) means.
- (c) Justify the claim in line (4).
- (d) Why do we know that (x_{n_k}) converges, as claimed in line (5)?
- (e) Why do we know that (x_n) converges, as claimed in line (6)?

7. You have now seen that, if A and B are complete subsets of a metric space, then both $A \cap B$ and $A \cup B$ are also complete. Only one of these two results can be extended to the case with arbitrarily many subsets instead of only two. Which can be extended? Provide a proof for the one that can be extended and a counter-example for the one that cannot be extended.